



جامعة ٦ أكتوبر
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علوم الحاسب ونظم المعلومات
Info Sys & Comp Science

Practical Project

Assistive Disability Interaction System

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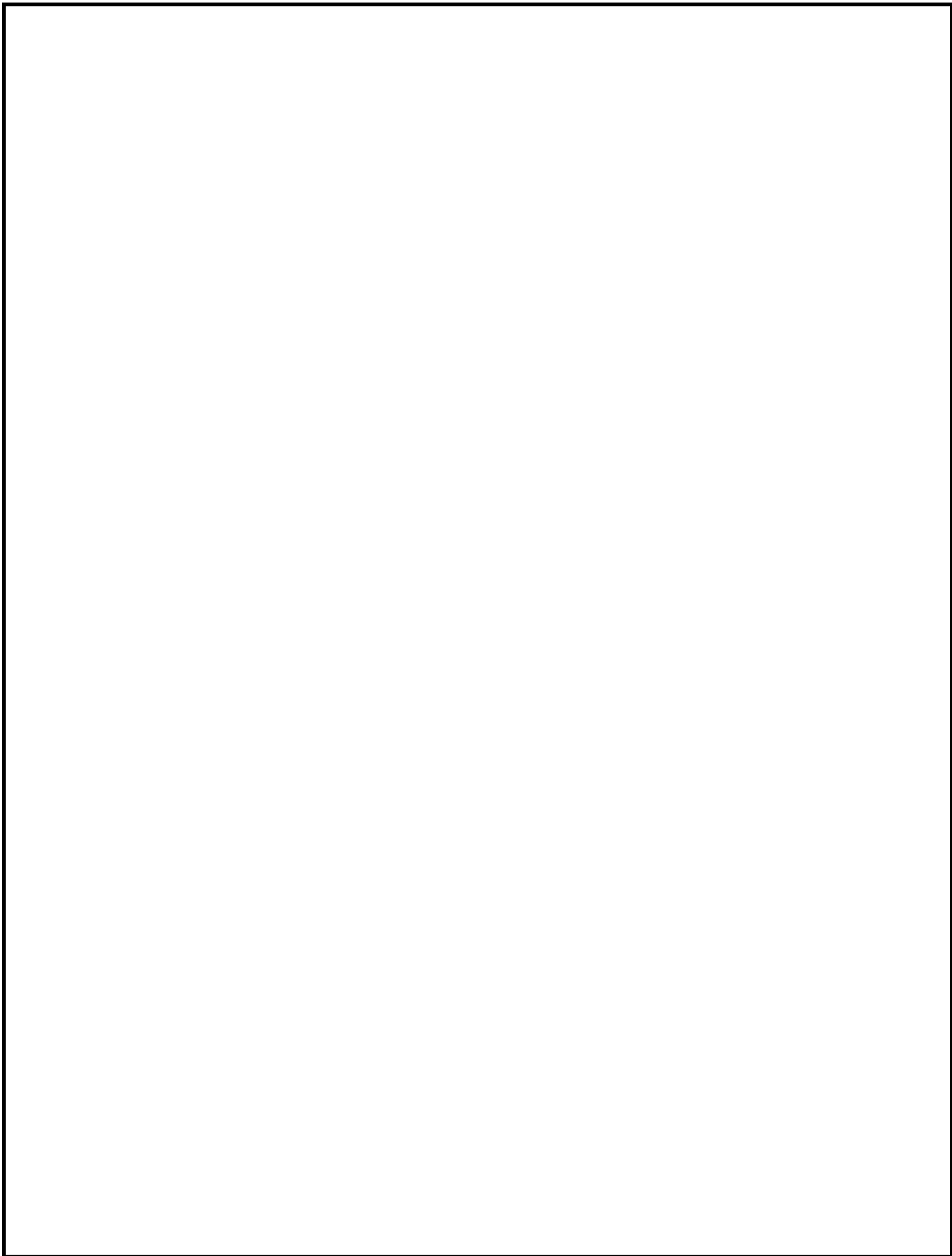
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Chapter Two

Design and Methodology

2.1 Requirements and Analysis

2.1.1 User requirements

2.1.2 Software requirements

2.1.2.1 Functional requirements

2.1.2.2 Non-functional requirements

2.1.1 Software Process Model:

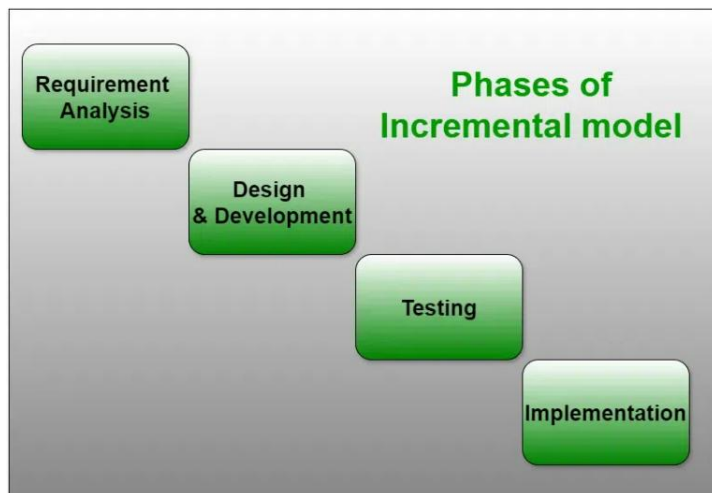
The system design model that we use is Incremental model according to system needs and our team suggestions

2.1.1 What is Incremental Model?

The **Incremental Model** is a software development approach where the system is built and delivered in small, functional parts called **increments**. Each increment adds more features to the system until the complete system is delivered. Early increments may contain core functionality, and subsequent increments enhance the system with additional features.

2.1.2 Phases of Incremental Model

- Requirement
- Design
- Testing phases
- Implementation phase



2.1.3 Advantages of Incremental Model

- Early delivery of core functionality allows users to start using the system sooner
- It is easier to manage risk and make changes in later increments.
- Allows feedback from users after each increment to improve subsequent parts.

2.1.4 Disadvantages of Incremental Model

- Incomplete systems in early increments may not provide full functionality to users.
- Requires careful planning to ensure all increments integrate smoothly.
- Can become costly if initial design is not flexible enough for changes.

2.2 System Models

2.2.1 Interaction perspective

2.2.1.1 Use Case Diagram

2.2.1.2 Sequence Diagram

2.2.2 Data Flow Diagram

2.2.2.1 Context Diagram

2.2.2.2 DFD (Level 1)

2.2.2.3 DFD (Level 2)

2.2.3 Activity Diagram

2.2.4 Structural perspective

2.2.4.1 Class Diagram

2.2.5 Entity Relationship Diagram (ERD)

References

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